



SAP S/4HANA
Maintenance Service

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English

Maintenance Service for Utilities

Cookbook

SAP S/4HANA 2023

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1 Introduction

The existing Customer Service (CS) integration of utilities maintenance processes in the component [Work Management](#) (WM) will be replaced with the standard SAP S/4HANA service using the maintenance service order (MSO) in combination with the plant maintenance order (PM order), see the Help Portal documentation:

[Work Management in SAP S/4HANA Service](#)

This integration requires some additional system setup activities, which are described in the following chapters. Please also check [SAP Note 3246547 - ISU Integration with S/4HANA Service](#) and the related notes.

Restrictions in SAP S/4HANA 2022:

Please check [SAP Note 3209741 - SAP S/4HANA 2022: Release information and restrictions for Maintenance Service in SAP S/4HANA](#) for further detailed information and restrictions for SAP S/4HANA Maintenance Service.

Restrictions in SAP S/4HANA 2023:

Please check [SAP Note 3330419 - This SAP Note informs you about restrictions related to SAP S/4HANA Service](#) regarding restrictions in SAP S/4HANA 2023.

2 Financial Accounting

2.1 FI-CA as Open Item Accounting

Please note that the integration of SAP S/4HANA Maintenance Service Order with FI-CA is not supported for S/4HANA 2022. For SAP S/4HANA 2023 FPS00, this integration is supported, but restrictions still apply. Please check the links in Chapter 1 [Introduction](#).

In the IS-U system, FI-CA is used as the subledger for open item accounting, and is also required for service transactions.

Activate it under: *IMG > Service > Master Data > Contract Account > Define Basic Settings*.

2.2 Item-Based Accounting for Service Management

During maintenance service (MS), the PM order is used as the CO object for the MS transaction items. This requires item-based accounting to be used.

Activate it under *IMG > Service > Transactions > Settings for Service Transactions > Integration > Enable Item-Based Accounting for Service Management*.

3 Organizational Model

3.1 Set Up Integration Between Business Partners and Organizational Units

The settings in this activity are required to enable an automated creation of business partners for HR employees and organizational units (HRALX), and to enable cost center maintenance in the employee master data (PLOGI).

1. Log on to the SAP S/4HANA system as a configuration user and choose the following navigation path: SPRO > *Service* → *Master Data* → *Business Partner* → *Integration Business Partner-Organization Management* → *Set Up Integration with Organizational Management*
2. Maintain the following settings:

Group	Sem.Abb.	Value Abbr	Description
HRALX	HRAC	X	Activate HR Integration
HRALX	OBPON	ON	Integration O-BP Activated
HRALX	ONUMB	1	Business Partner Number Assignment (Org. Unit)
HRALX	OSUBG	<empty>	Business Partner Subgroup (Organizational Unit)
HRALX	PBPON	ON	
PLOGI	ORGA	<empty>	Integration Switch: Organizational Assignment

3.2 Maintaining Organizational Units

The MS transactions require organizational units for sales and service, to derive the sales area, service organization and service team in the transaction.

3.2.1 Organizational Unit for Sales

Create the sales organizational unit with the following details:

- Functions
 - o Sales Organization: *selected*
- Allow Org. Unit to be Determined
 - o Sales: *selected*
- Attributes / Scenario Sales
 - o Division: *<your value>*
 - o Country: *<your value>*
 - o Ref. currency for document: *<your value>*
 - o Distribution channel: *<your value>*
 - o R/3 sales organization: *<your value>*

3.2.2 Organizational Units for Service

3.2.2.1 Service Organization

The planning plant in the S/4HANA PM order is determined from the “R/3 maintenance planning plant” attribute for the service organizational unit.

Create the service organizational unit with the following details:

- Functions
 - o Service Organization: *selected*
- Allow Org. Unit to be Determined
 - o Service: *selected*

- Attributes / Scenario Service
 - o R/3 maintenance planning plant: *<your value>*
 - o Postal code: *<your value>*

3.2.2.2 Service Team

The service team is determined using the organizational data determination rule assigned to the access sequence for the partner function 00000056 "Service Employee Group" in the partner determination procedure for the MS item.

The work center and plant in the S/4HANA PM order are determined from the attribute "WORKCENTER" for the service team organizational unit.

Create the service team organizational unit with the following details:

- Functions
 - o Service Team: *selected*
- Allow Org. Unit to be Determined
 - o Service: *selected*
- Attributes / Scenario Service
 - o Postal code: *<your value>*
 - o WORKCENTER: *<your value>*

3.3 Organizational Structure Check

3.3.1 Updating List of Organizational Units

You can use this program to update the organizational units display manually. This then enables you to see the organizational units that you have created straightaway in the search help.

If you want to create a new organizational unit or position for the organizational model, you will not be able to see this until the following day. It will not, for example, be displayed immediately in the search help for sales areas in the business partner master.

Execute this for the object type 'O' Organizational unit.

▼	Service
>	Cross-Application Components
>	UI Framework
>	Service-Specific Settings for UI Framework
>	Industry-Specific Solutions
>	Basic Functions
▼	Master Data
▼ 	Organizational Management
>	Number Range Maintenance
> 	Organizational Data Determination
▼ 	Tools
 	Organizational Management Buffer Setting
 	Update List of Organizational Objects

3.3.2 Checking Settings for Attribute Maintenance

You can use this report to check the consistency of the attribute assignments with the organizational units.

✓	Service
>	Cross-Application Components
>	UI Framework
>	Service-Specific Settings for UI Framework
>	Industry-Specific Solutions
>	Basic Functions
✓	Master Data
✓ 	Organizational Management
>	Number Range Maintenance
> 	Organizational Data Determination
✓ 	Tools
 	Organizational Management Buffer Setting
 	Update List of Organizational Objects
 	Check Settings for Attribute Maintenance

3.3.3 Checking Organizational Data Settings

This activity executes the *Check Settings for Organizational Data* program to compare the sales area data defined in *Sales and Distribution (SD)* and in SAP S/4HANA Service. You can also check the Customizing settings for both specific transaction types, and the organizational model data, using various search criteria.

✓	Service
>	Cross-Application Components
>	UI Framework
>	Service-Specific Settings for UI Framework
>	Industry-Specific Solutions
>	Basic Functions
✓	Master Data
✓ 	Organizational Management
>	Number Range Maintenance
> 	Organizational Data Determination
✓ 	Tools
 	Organizational Management Buffer Setting
 	Update List of Organizational Objects
 	Check Settings for Attribute Maintenance
 	Translate Organizational Units
 	Organizational Data Settings Check

3.3.4 Checking Integration Between Business Partners and Organizational Units

This report runs a consistency check for the integration between business partner data and organizational units. If not all data is available for the business partner, you can synchronize and repair the data first.

Execute the report. Select 'Organizational Unit(s)' and leave the value empty (=active). If the report shows errors or inconsistencies, choose *Repair*.

▼	Service
>	Cross-Application Components
>	UI Framework
>	Service-Specific Settings for UI Framework
>	Industry-Specific Solutions
>	Basic Functions
▼	Master Data
>	Organizational Management
>	Service Teams
▼	Business Partner
	Configure Business Partner
>	Basic Settings
	Define Account Classification
	Define Basic Settings for Account Life Cycle
	Activate Contact Creation with Contact Person Role
▼	Integration Business Partner-Organization Management
	Set Up Integration with Organizational Management
	Create Business Partner Initially

3.3.5 Updating Attribute Buffer for Generic Attributes

Sales areas are calculated on demand when these are required by an S/4 service transaction or a business partner. This is an expensive process. To speed up this process, the sales areas are buffered.

The `HRBCI_ATTRIBUTES_BUFFER_UPDATE` report is used to buffer the attributes for organizational management objects.

3.4 Organizational Data Profile

The standard profile 0000000000008 / scenarios SALE and SERVICE is used. It contains the following determination rules:

- Scenario SALE – rule 10000144
 - Determination of the responsible sales organizational unit from the sales areas maintained in the business partner (“sales areas incl. master data”)
- Scenario SERVICE – rule 10000276:
 - Determination of the responsible service organizational unit according to the partner postal code
 - Service organization: selected
 - The planning plant is derived from the service organization determined
 - The work center is derived from the service team determined (determination rule in the partner determination procedure)

4 HCM Integration with Business Partners

A business partner employee (BP role BUP003 Employee) is required in the MS transactions for the partner function 00000013 Person Responsible. It is determined from the reference objects in the MSO header (functional location and/or equipment), to which the personnel number of one or more HCM employees can be assigned.

The partner function 00000013 Person Responsible is not mandatory in the MS transaction. This is required to align it with the settings for the partner function VW Person Responsible in the PM order.

For details please see the blog [PPM – HCM Integration with Business Partners](#) and [SAP Note 2792166 - Same number for BP and employee](#).

5 Partner Processing

5.1 Partner Determination Procedure

5.1.1 Header

The standard procedure 00000011 "Maint.Service Order" is used for the header.

Align it with the Utilities processes by changing the access sequence for the partner function 00000001 "Sold-to Party (CRM)" to IUEQ "Preceding Document: Activity Partner -> Preceding Document: Sold-To Party -> IS-U Device: Sold-To Party".

5.1.2 Item

The standard procedure 00000012 "Maint. Service Order (Item)" is used for the item.

Align it with the Utilities processes by changing the access sequence for the partner function 00000001 "Sold-to Party (CRM)" to IUEQ "Preceding Document: Activity Partner -> Preceding Document: Sold-To Party -> IS-U Device: Sold-To Party".

5.2 Map Partner Functions

Check that the service partner functions in the header and item partner determination procedures are mapped to the corresponding SD partner functions.

✓	Service
>	Cross-Application Components
>	UI Framework
>	Service-Specific Settings for UI Framework
>	Industry-Specific Solutions
✓	Basic Functions
>	Content Management
>	Enterprise Organizational Management
✓ 	Partner Processing
 	Define Partner Functions
 	Define Access Sequences
> 	Partner Teams
 	Define Partner Determination Procedure
 	Map Partner Functions

6 Pricing

6.1 Number Ranges for Document Conditions in CRM Documents

Number range '01' is required for the document conditions (number range object CRM_KNUMV).

Use transaction SNUM to create number range 01.

6.2 Define Condition Types

6.2.1 Planned Costs from PM Order

The pricing procedure for the maintenance service quotation needs to have a condition type to which the planned and actual costs are sent:

- Planned costs in the resource-related quotation are sent to the MS quotation
- Actual costs in the resource-related billing are sent to the debit memo request

Create a condition type with the following settings:

Condition Type: Charged costs	Access Sequence:
Records	
Control Data 1	
Condition Class: * B Prices Calculation Type: * C Quantity Condition Category: Rounding Rule: Commercial Structure Condition: None Condition Function: Dat.Rec.Source: Condition Technique (SAP S/4HANA) ▼ Inclusion Point: ☐ Plus/Minus: positive and negative	
Group Condition	
Group Condition: ☐ RoundDiffComp: ☐ Group Cond. Routine:	
Changes which can be made	
Manual Entries: C Manual entry has priority Header Condition: ☐ Item Condition: ☒ Delete: ☒ Amount/Percent: ☒ Quantity Relation: ☐ Value: ☐ Calculation Type: ☐	
Master Data	
Proposed Valid From: Today's date Proposed Valid To: 31.12.9999 Pricing Procedure: Delete from DB: B without popup Condition Update: ☐	

Master Data	
Proposed Valid From: ☐ Today's date	Pricing Procedure:
Proposed Valid To: ☐ 31.12.9999	Delete from DB: without popup
	Condition Update: ☐
Scales	
Scale Base Type: ☐	Scale Routine:
Check Scale: ☐ None	Scale Unit:
Scale Type: ☐ can be maintained in condition record	
Control Data 2	
Currency Conversion: ☐	Exclusion: ☐
	Pricing Date: ☐ Standard (KOMK-PRS
Accruals: ☐	
Used for Var.Config.: ☐	ReL. for Acct Assigt: ☐ Relevant for account a
Invoice List Cond.: ☐	Enabled for CPF: ☐
Quantity Conversion: ☒	
Intercomp.Billing: ☐	ServChgeSettlem: ☐
	Zero Value Proc.: ☐
Text Determination	
TextDermProcedure:	Text ID:

6.2.2 Mark-Up Percentage for Planned Costs

You can create a percentage condition, to adjust the planned cost in the MS quotation with the overhead.

6.3 Set Pricing Procedures

Create a pricing procedure which includes the following conditions in this order:

1. Product price, for example condition PR00
2. [Planned Costs from PM Order](#)
3. [Mark-up Percentage for Planned Costs](#)

7 Products

Two types of products are used in the MS transactions:

- [MS Products](#)
 - o These are the service products in the MS order item
 - o They have a price, which is used if the item is quoted or billed with a fixed price
- [Resource-Related Products](#)
 - o Materials for the planned costs in the MS quotation and MS order and actual costs in the debit memo request for resource-related billing
 - o Dedicated materials are available for services (working hours), components (spare parts) and expenses

7.1 Item Category Group

Use the following standard item category groups delivered:

- DIEN “Service w/ Delivery” for the [MS Products](#)
- LEIS “Service w/o Delivery” for the service items within [Resource-Related Products](#)
- NORM “Standard item” for the component items within [Resource-Related Products](#)

7.2 Maintenance Service Products

Create a product representing for example the price for the construction of a new connection. Use the material type SERV “Service Materials”.

7.3 Resource-Related Products

Create products for

- Service hours used in the PM order operations
- Spare parts used in the PM order components

8 Maintenance Service Transactions

8.1 Profit Center

You must enter the profit center for both the MS quotation and the order items. It is synchronized bi-directionally with the PM order created for the MS order.

You can still change the profit center after the maintenance order has been created. However, this is no longer possible once a cost booking has been made to the MS order item.

The profit center can be entered manually or defaulted in the following sequence:

1. Manual entry
2. Substitution rule
3. Person responsible cost center
4. Default profit center assigned to the controlling area:

Please adapt your system according to these options.

8.2 Maintenance Service Quotation

MS quotation transactions are used in the resource-related quotation (RRQ) process. The quotation receives planned costs from the PM order created for the MS quotation.

8.2.1 Transaction Type

A single MS transaction type can be used to manage all Utilities MS quotation processes.

Use the standard transaction type delivered MSQ1 “Maint. Srv Quotation”.

8.2.2 Item Category

In the Utilities solution, the item category represents the various business processes, such as new connection or metering operations.

The following standard item categories delivered are used:

- Item / main item (if the resource-related quotation is used)
 - MSQ1 “Maint. Item (Quote)” for the [Maintenance Service Products](#)
 - The item object type is BUS2000192 “Service Maintenance Item”
- Sub-item (if the resource-related quotation is used)
 - MSQP “Service Item (Maint)” for the service item products for [Resource-Related Products](#)
 - The item object type is BUS2000140 “Service Product Item”
 - MSQM “Spare Item (Maint.)” for the spare item products of [Resource-Related Products](#)
 - The item object type is BUS2000146 “Service Material Item”

8.2.3 Item Category Determination

Use the standard determinations delivered:

Transaction Type	Item Category Group	Main Item Category	Item category	Comment
MSQ1	DIEN		MSQ1	Main Item: Maintenance Service Item
MSQ1	DIEN	MSQ1	MSQP	Sub Item: Service Product Item (hours)
MSQ1	NORM	MSQ1	MSQM	Sub Item: Service Material Item (spares)

8.2.4 Copy Control

When the MS quotation is accepted by the customer, a follow-up MS order is created from the MS quotation.

Use the standard copy controls delivered:

- Copying control for transaction types
 - MSQ1 to MSO1
- Item category determination when copying
 - MSQ1 + MSO1 = MSI1
- Copying control for item categories
 - MSQ1 to MSI1

8.3 Maintenance Service Order

8.3.1 Transaction Type

A single MS transaction type can be used to manage all Utilities MS order processes.

Use the standard transaction type delivered MSO1 “Maint. Service Order”.

8.3.2 Item Category

In the Utilities solution, the item category represents the various business processes, such as new connection or metering operations.

Use the standard item category delivered MSI1 “Maint. Service Item” for the [Maintenance Service Products](#). The item object type is BUS2000192 “Service Maintenance Item”.

8.3.3 Item Category Determination

Use the standard determination delivered:

Transaction Type	Item Category Group	Main Item Category	Item Category
MSO1	DIEN		MSI1

8.4 Plant Maintenance Integration

Integration with the MS order and PM order includes two activities:

- Map Order Types
- Map Priorities

✓	Service
>	Cross-Application Components
>	UI Framework
>	Service-Specific Settings for UI Framework
>	Industry-Specific Solutions
>	Basic Functions
>	Master Data
>	Integration with Other SAP Solutions
✓ 	Transactions
>	Basic Settings
> 	Settings for Activities
> 	Settings for Opportunities
> 	Settings for Leads
>	Settings for Solutions
>	Settings for Contracts
✓	Settings for Service Transactions
 	Define Valuation Types
 	Define Service Types
 	Define Availability and Response Times
 	Define Accounting Indicators
	SLA Escalation Management
 	Define Profile for Warranty Check
	Settings for Automatic Credit Check
	Settings for Rule-Based Dispatching
 	Define Reasons For Rejection
>	Prediction of Service Parts and Service Duration
>	Questionnaires for Service Processes and Service Confirmations
✓	Integration
 	Replicate Service Characteristics Relevant to Controlling
 	Enable Item-Based Accounting for Service Management
	Set Up Time Sheet and Controlling Integration
✓	Plant Maintenance Integration
 	Map Order Types
 	Map Priorities

8.4.1 Map Order Types

In this step, you can map the transaction type and item category for a MS transaction to a PM order type. This mapping is used to determine a valid maintenance order type for a maintenance service order along with an item category and vice versa. A default rejection reason for a maintenance service order item is provided automatically. This can be used when you set the corresponding maintenance order to **Do not execute**.

Use the following standard mappings delivered:

- MSO1 + MSI1 = PM01
- MSQ1 + MSQ1 = PM01

8.4.1 Map Priorities

In this step, the service priorities can be mapped to plant maintenance priorities. This mapping is used in integration scenarios, such as the creation of maintenance orders from maintenance service orders and vice-versa.

The standard delivery includes mappings for all standard priorities.

9 Resource-Related Quotation

When the price of the MS quotation is calculated based on the planned costs of the PM order, the dynamic item processor (DIP) profile is used to determine the materials used in the MS quotation sub-items. The DIP profile is assigned to the main item in the MS quotation.

Use the standard DIP profile delivered MS000001 "Profile for Maintenance Service".

Adapt usage 11 "Quotation creation and sales pricing" for your products used in MS:

- Determine the materials for service hours and components from [Resource-Related Products](#) with criteria such as activity type or cost element
- For "Transfer Quantity/Costs" select "Transfer Costs and Quantity" for all materials

10 Resource-Related Billing

When the billing relevance of the MS order Item is "Itemized Billing" or "Summarized Billing", resource-related billing is used to create a debit memo request (DMR) with the Fiori app or transaction DP92. The DIP profile is used to create the DMR.

Use the standard DIP profile delivered MS000001 "Profile for Maintenance Service". It uses the standard sales document type DR1 "Srv. Debit Memo Rq" for the DMR.

Adapt usage 1 "Billing and results analysis" for your products used in MS:

- Determine the materials for service hours and components from [Resource-Related Products](#) with criteria such as the activity type or cost element
- For "Transfer Quantity/Costs" select "Transfer Costs and Quantity" for all materials

10.1 Assign Sales Area to Sales Document Types

Maintain the activities in this Customizing activity according to your sales areas.

1. Combine Sales Organizations
2. Combine Distribution Channels
3. Combine Divisions
4. Assign Sales Document Types Permitted for Sales Areas

✓ 	Sales and Distribution
> 	Master Data
> 	Basic Functions
✓ 	Sales
✓ 	Sales Documents
✓ 	Sales Document Header
	 Define Sales Document Types
	 Maintain Sales Document Types for Fashion Management
	 Maintain Sales Document Types per Sales Area for Fashion Management
	 Assign Custom Routines to Sales Document Types
	 Convert Language For Each Sales Document Type
	 Maintain Sales Document Types per Sales Area for Supply Assignment (ARun)
	 Define Number Range Prefixes for SD Documents
	 Define Number Ranges for Sales Documents
	 Define Purchase Order Types for Sales Documents
	 Define Order Reasons
	 Assign Sales Document Types and Sales Organizations to Order Reasons
	 Define Usage Indicators
	 Assign Sales Area to Sales Document Types

10.2 Assign Conditions to Sales Document Types

The planned and actual costs from the PM order are entered in this condition type in the MSQ sub-items (RRQ) and in the DMR sub-items (RRB).

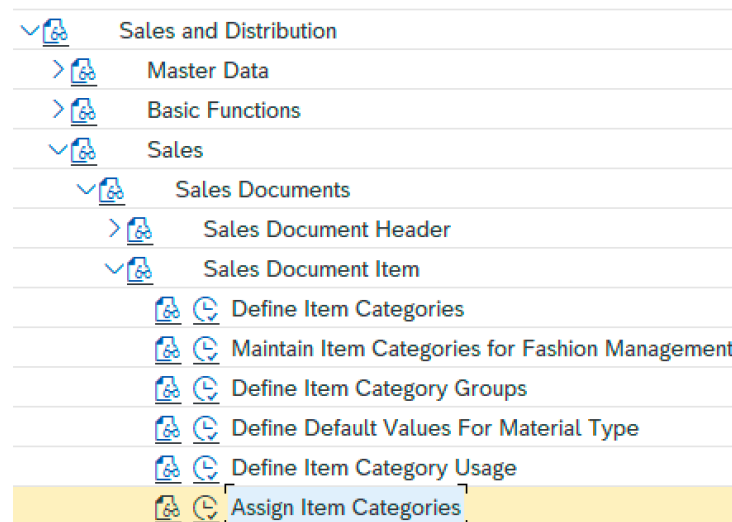
Assign your condition type from [Planned Costs from PM Order](#) to the sales document type for the DMR (DR1 as specified in the DIP profile).

✓ 	Plant Maintenance and Customer Service
>	Warranty Claim Management
> 	Master Data in Plant Maintenance and Customer Service
> 	Maintenance Plans, Work Centers, Task Lists and PRTs
✓ 	Maintenance and Service Processing
✓ 	Basic Settings
	 Maintain Authorizations for Processing
	 Planning of Background Jobs for PDC
	 Define Download Parameters for PDC
> 	General Data
> 	Print Control
> 	General Order Settlement
> 	Settings for Display of Costs
✓ 	Quotation Creation and Billing for Service Orders
	 Profiles for Quotation Creation, Billing, Results Analysis
	 Check Profile Settings
	 Define Apportionment Reasons
	 Assign Conditions to Sales Document Types

10.3 Assign Item Categories

The item categories for the items in the DR1 debit memo request created by resource-related billing are determined from this activity.

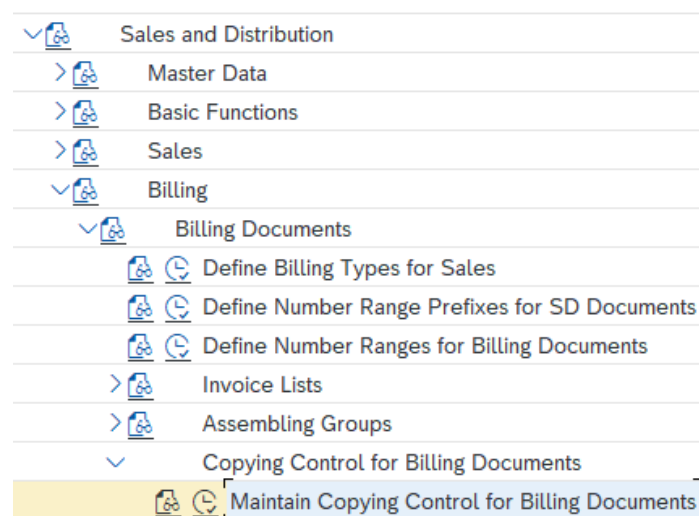
Use the standard determinations delivered for the sales document type DR1.



10.4 Define Copying Control for Sales Document to Billing Document

When the billing document is created from the debit memo request, you must maintain copy control from the sales document type to the billing document.

Use the standard copying control delivered from the source sales document type DR1 to the target billing type L2.



Activities

Perf...	Name of Activity
	Define Copying Control for Sales Document to Billing Document
	Define Copying Control for Billing Document to Billing Document
	Define Copying Control for Delivery Document to Billing Document

11 Fixed Price Billing

When the billing relevance of the MS order item is “Billing on Completion”, service billing is used to create a billing document request (BDR) with the Release for Billing app.

This is used for fixed price billing, when the MS order item is billed with the condition amount for the MS product (catalog price).

11.1 Sales Document Type

The standard sales document type delivered BSVO “Service Order eBDR” is used for the billing document request.

11.2 Billing Type

The standard billing type delivered SVOB “Cust. Mgmt Service Order BDR” is used for the billing document request.

11.3 Assign Service Transaction Types to Billing Document Request Types

When the MS order is released for billing, a billing document request is created with the sales document type and billing type assigned to the transaction type for the MS order.

The item category in the BDR is determined by mapping the item categories in the MS order to the target item categories in the BDR.

Use the standard mapping delivered:

- Process Type Mapping
 - o The MS order transaction type MSO1 is mapped to the sales document type BSVO and billing type SVOB
- Item Category Mapping
 - o The item category MSI1 is mapped to item category CS01

▼	Service
>	Cross-Application Components
>	UI Framework
>	Service-Specific Settings for UI Framework
>	Industry-Specific Solutions
▼	Basic Functions
>	Content Management
>	Enterprise Organizational Management
>	Partner Processing
>	Pricing
▼	Billing
	Configure Billing
	Assign Service Transaction Types to Billing Document Request Types

11.4 Define Copying Control for Sales Document to Billing Document

When the billing document request is created from the MS order using the Release for Billing app, you must maintain copy control for the sales document to the billing document.

Use the standard copying control delivered from the source sales document type BSVO to the target billing type SVOB.

▼	Sales and Distribution
>	Master Data
>	Basic Functions
>	Sales
▼	Billing
▼	Billing Documents
	Define Billing Types for Sales
	Define Number Range Prefixes for SD Documents
	Define Number Ranges for Billing Documents
>	Invoice Lists
>	Assembling Groups
	Maintain Copying Control for Billing Documents

Activities

Perf...	Name of Activity
✓	Define Copying Control for Sales Document to Billing Document
	Define Copying Control for Billing Document to Billing Document
	Define Copying Control for Delivery Document to Billing Document

11.5 Define Copying Control for Billing Document to Billing Document

When the billing document is created from the billing document request, you must maintain copy control from the billing document to the billing document .

Use the standard copying control delivered from the source billing type SVOB to the target billing type F2.

✓	Sales and Distribution
>	Master Data
>	Basic Functions
>	Sales
✓	Billing
✓	Billing Documents
	Define Billing Types for Sales
	Define Number Range Prefixes for SD Documents
	Define Number Ranges for Billing Documents
>	Invoice Lists
>	Assembling Groups
	Maintain Copying Control for Billing Documents

Activities

Perf...	Name of Activity
	Define Copying Control for Sales Document to Billing Document
✓	Define Copying Control for Billing Document to Billing Document
	Define Copying Control for Delivery Document to Billing Document

12 Plant Maintenance

PM configuration is performed as usual. No specific Customizing related to the MSO is required.

13 Work Management

13.1 Define Service Objects

Define service objects where the parameters that are automatically transferred into the order are specified.

☐		SAP Utilities
☐		Kundenbeziehungen
☐		Basic Settings/Enterprise Structure
☐		Basic Functions
☐		Master Data
☐		Device Management
☐		Energy Data Management
☐		Contract Billing
☐		Invoicing
☐		Customer Service
☐		Work Management
☐		Master Data
☒		Define Service Objects
☐		Define Set Levels for Service Objects
☐		Allocate Service Objects to Division and Set Level
☐		Processing

Remarks about service object configuration:

- Create service objects for planning plants, which are responsible for those tasks.
- To be able to cover cases with MSO, service objects should be created with the ID "01"
- Service objects with the service object ID 01 are called service products.
Note: These service objects need a corresponding material master with the same name.
- Those service products shouldn't be flagged as configurable material.
- PMO types are created in PM Customizing steps
- The task list should be defined in advance.

Some examples of service objects are shown below:

PIPI	ID	Service Object	Ref. Type	Ref. Obj. Type
	01	APERIODIC METER READING	02	03
	01	APERIODIC METER READING FL	02	01
	01	DISCONNECT	02	01
	01	INSPECTION	01	02
	01	INSPECTION FL	01	01
	01	METER EXCHANGE	02	02
	01	METER EXCHANGE FL	02	01
	01	METER EXCHANGE GAS	02	02
	01	METER EXCHANGE GAS FL	02	01
	01	METER INSTALLATION	02	02
	01	METER INSTALLATION FL	02	01
	01	METER INSTALLATION GAS	02	02
	01	METER INSTALLATION GAS FL	02	01
	01	METER REMOVAL	02	02
	01	METER REMOVAL FL	02	01
	01	METER REMOVAL GAS	02	02
	01	METER REMOVAL GAS FL	02	01
	01	PERIODIC_REPLACEMENT	02	03

	01	PERIODIC_REPLACEMENT FL	02	01
	01	RECONNECT	02	01

13.2 Define Set Levels for Service Objects

Define set levels that are used to group service objects according to subject.

☐		SAP Utilities
☐	>	Kundenbeziehungen
☐	>	Basic Settings/Enterprise Structure
☐	>	Basic Functions
☐	>	Master Data
☐	>	Device Management
☐	>	Energy Data Management
☐	>	Contract Billing
☐	>	Invoicing
☐	>	Customer Service
☐	>	Work Management
☐	>	Master Data
☐		Define Service Objects
☒		Define Set Levels for Service Objects
☐		Allocate Service Objects to Division and Set Level
☐	>	Processing

The set level to group tasks for meter operations could be created as below:

Set Level	Description
0004	Meter operations

13.3 Allocate Service Objects to the Division and Set Level

Service objects are allocated to set levels. If a service object requires a specific order type or task list for specific divisions, create specific service objects for each of these and also allocate them to a division.

☐		SAP Utilities
☐	>	Kundenbeziehungen
☐	>	Basic Settings/Enterprise Structure
☐	>	Basic Functions
☐	>	Master Data
☐	>	Device Management
☐	>	Energy Data Management
☐	>	Contract Billing
☐	>	Invoicing
☐	>	Customer Service
☐	>	Work Management
☐	>	Master Data
☐		Define Service Objects
☐		Define Set Levels for Service Objects
☒		Allocate Service Objects to Division and Set Level
☐	>	Processing

Some examples of allocation are shown below:

ID	Service Object	Division	Set Level
01	METER EXCHANGE	01	0004
01	METER EXCHANGE FL	01	0004
01	METER EXCHANGE GAS	02	0004
01	METER EXCHANGE GAS FL	02	0004
01	METER INSTALLATION	01	0004
01	METER INSTALLATION FL	01	0004
01	METER INSTALLATION GAS	02	0004
01	METER INSTALLATION GAS FL	02	0004
01	METER REMOVAL	01	0004
01	METER REMOVAL FL	01	0004
01	METER REMOVAL GAS	02	0004
01	METER REMOVAL GAS FL	02	0004

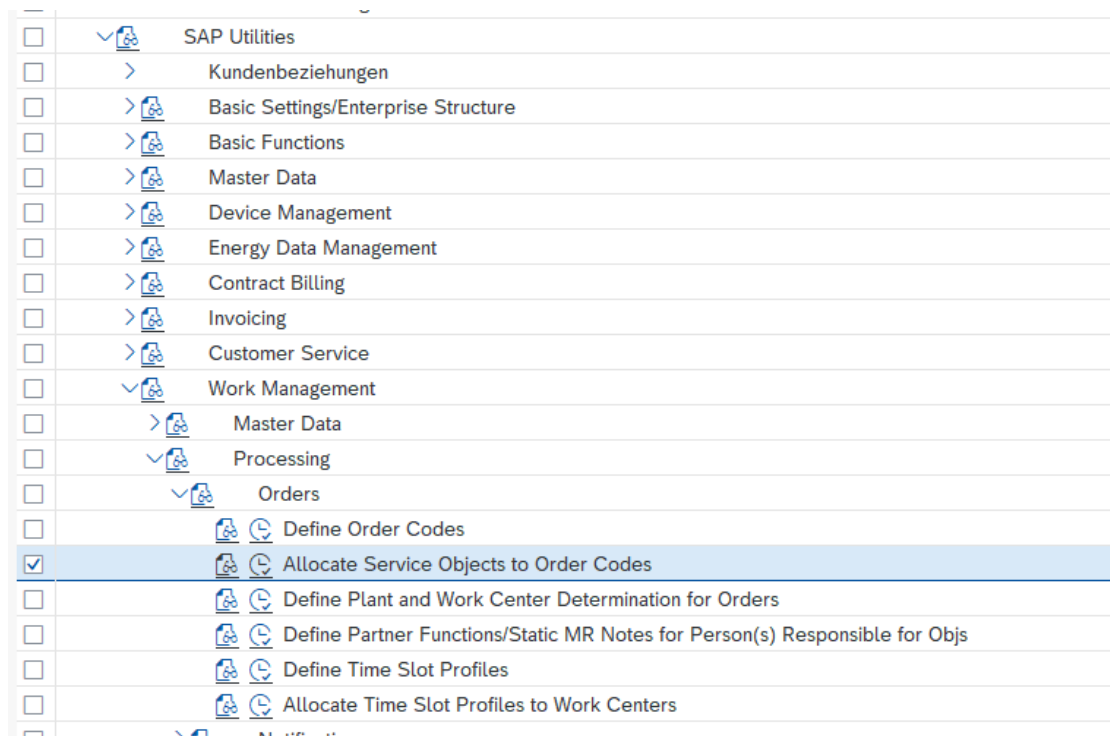
13.4 Define Order Codes

Default order codes are used here. If you need to change the order code manually or by using an enhancement for the transaction for which it was called, you can define custom order codes here.

☐		SAP Utilities
☐		Kundenbeziehungen
☐		Basic Settings/Enterprise Structure
☐		Basic Functions
☐		Master Data
☐		Device Management
☐		Energy Data Management
☐		Contract Billing
☐		Invoicing
☐		Customer Service
☐		Work Management
☐		Master Data
☐		Processing
☐		Orders
☒		Define Order Codes
☐		Allocate Service Objects to Order Codes
☐		Define Plant and Work Center Determination for Orders
☐		Define Partner Functions/Static MR Notes for Person(s) Responsible for Objs
☐		Define Time Slot Profiles
☐		Allocate Time Slot Profiles to Work Centers
☐		Notifications

13.5 Allocate Service Objects to Order Codes

Order codes mentioned in the previous step are maintained with the ID "01" and the relevant service object. ID "00" can be updated to create a PMO only without an MSO.



Here are some examples:

O. Code	Description	ID	Service Object
DC00	Disconnection order (collection)	01	DISCONNECT
INS2	Inspection order: Connection object	01	INSPECTION FL
INSP	Inspection order: Equipment	01	INSPECTION
PRE1	Periodic repl. order: Device location	01	PERIODIC_REPLACEMENT FL
PRE2	Periodic repl. order for connection obj.	01	PERIODIC_REPLACEMENT FL
PREP	Order for periodic replacement device	01	PERIODIC_REPLACEMENT
RC00	Reconnection order (collection)	01	RECONNECT
SMR1	Order for aper. MR for functional loc.	01	APERIODIC METER READING FL
SMR2	Order for aper. MR for connection object	01	APERIODIC METER READING FL
SMRO	Order for aperiodic MR for device	01	APERIODIC METER READING

13.6 Define Plant and Work Center Determination for Orders

Assign the regional structure area and planning plant to the IDs defined above and service objects. Then assign the work center to the correct regional structure group for each of these service objects.

☐		SAP Utilities
☐	>	Kundenbeziehungen
☐	> 	Basic Settings/Enterprise Structure
☐	> 	Basic Functions
☐	> 	Master Data
☐	> 	Device Management
☐	> 	Energy Data Management
☐	> 	Contract Billing
☐	> 	Invoicing
☐	> 	Customer Service
☐		Work Management
☐	> 	Master Data
☐		Processing
☐		Orders
☐	 	Define Order Codes
☐	 	Allocate Service Objects to Order Codes
☒	 	Define Plant and Work Center Determination for Orders
☐	 	Define Partner Functions/Static MR Notes for Person(s) Responsible for Objs
☐	 	Define Time Slot Profiles
☐	 	Allocate Time Slot Profiles to Work Centers
☐	> 	Notifications